Lab 3 FDSAI

2026-2027

1. Using the following table:

Day	ScreenTime (hours)
Mo	5
Tu	6
We	4
Th	7
Fr	8
Sa	10
Su	9

a) Create a line chart showing screen time over the week.
b) Add:
    a. Title, axis, labels, and clear units.
    b. A grid (optional – test if it helps or distracts).
c) Make a second version as a bar chart and discuss:
    a. Which chart communicates the pattern more clearly?

2. Plot some functions (taking into consideration the best practices):

$$f_1(x) = max(0, x)$$

$$f_2(x) = \frac{e^{2x} - 1}{e^{2x} + 1}$$

$$f_3(x) = \frac{1}{1 + e^{-x}}$$

$$f_4(x) = \frac{1}{\mu^2 + 1} * e^{-\frac{(x-\mu^2)}{\sigma^2}}$$

$$f_5(x) = 0.5 * x * (1 + tanh(0.797885x + 0.035677x^3)$$

3. Using the following table:

Student	Score
1	70
2	75
3	78

4	82
5	74
6	90
7	92
8	95
9	60
10	100

a) Create a boxplot to identify possible outliers.
b) Highlight outliers in a scatter plot as separate color points.
c) How would misleading axis scales make outliers seem „normal”?